ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

USING DEEP LEARNING FOR GRAMMATICAL ERROR CORRECTION IN AFAAN OROMO

A MASTER'S THESIS

BY

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DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

SCHOOL OF POST GRADUATE STUDIES

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A Thesis Submitted as a Partial Fulfillment to the Requirements for the Award of the Degree of Master of Science in Electrical and Computer Engineering (Computer Engineering)

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DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

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DECLARATION

I hereby declare that this thesis entitled "**Using Deep Learning for Grammatical Error Correction in Afaan Oromo**" is my own work with the guidance of my advisor. This thesis has not been submitted to any university for a similar purpose. The references used in this proposal are duly recognized by proper citations.

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APPROVAL PAGE

This is to certify that the thesis prepared by **Mr. Solen Abera Deressa** entitled "**Using Deep Learning for Grammatical Error Correction in Afaan Oromo"** and submitted as a partial fulfillment for the award of the Degree of Master of Science in Electrical and Computer Engineering (Computer Engineering) complies with the regulations of the university and meets the accepted standards with respect to originality, content and quality.

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ABSTRACT

Grammatical errors can greatly hinder the clarity and effectiveness of communication. This results in decreased reader engagement and prevents the intended message from being conveyed effectively. However, existing Deep Learning (DL) models for Grammatical Error Correction (GEC) often suffer from data scarcity. These models also lack adaptation to the unique linguistic features of under-resourced languages. To tackle these, the thesis focuses on leveraging DL techniques to develop advanced language processing models tailored for Afaan Oromo GEC. A dataset of 9,500 parallel Afaan Oromo sentences, each paired with its grammatically incorrect and corrected versions, is prepared and used to train and evaluate the performance of Text-to-Text Transfer Transformer (T5) and Bidirectional and Auto-Regressive Transformers (BART) models. The results indicate that the T5 and BART models were highly effective in Afaan Oromo GEC. The BART model achieved a superior $F_{0.5}$ score of 0.8144 compared to 0.8021 for the T5 model. These results demonstrate a performance improvement over the previous Afaan Oromo hybrid grammar checker, with a 3.09% increase for BART and a 1.52% increase for T5. The demonstrated improvements in GEC will facilitate clearer and more effective communication for both native speakers and learners of Afaan Oromo language.

Keywords: GEC, Deep Learning, T5, BART, Natural Language Processing, Afaan Oromo.

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LIST OF ACRONYMS AND ABBREVIATION

BART	Bidirectional and Auto-Regressive Transformers
BEA-2019	Building Educational Applications 2019 Shared Task
BERT	Bidirectional Encoder Representations from Transformers
CLM	Causal Language Modeling
DL	Deep Learning
ESD	Erroneous Span Detection
ESC	Erroneous Span Correction
GEC	Grammatical Error Correction
GLEU	Generalized Language Evaluation Understanding
GLMs	Generative Language Models
GPT	Generative Pre-trained Transformer
GPUs	Graphics Processing Units
JFLEG	JHU FLuency-Extended GUG corpus
LANG-8	Language-8
LM	Language Model
LSTM	Long Short-Term Memory
M^2	Max-Match
MLM	Masked Language Modeling
NLP	Natural Language Processing
NMT	Neural Machine Translation
NN	Neural Network
NLTK	Natural Language Toolkit
NSP	Next Sentence Prediction
PLMs	Pre-trained Language Models
RAM	Random Access Memory
RMSProp	Root Mean Square Propagation
RNN	Recurrent Neural Network
SGD	Stochastic Gradient Descent
SMT	Statistical Machine Translation
SOV	Subject-Object-Verb
T5	Text-to-Text Transfer Transformer
WOEs	Word Ordering Errors

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CHAPTER ONE

INTRODUCTION

1.1 Background

Writing is a skill that is acquired through learning. It is particularly difficult for individuals who are not native speakers of a language. Although even native speakers sometimes make errors in punctuation, spelling, and word choice, non-native writers often face additional challenges[1].

Afaan Oromo, which belongs to the Cushitic branch of the Afroasiatic language family, is the most widely spoken language in Ethiopia[2]. Despite its widespread use, Afaan Oromo faces challenges in terms of language technology development, particularly in the area of Grammatical Error Correction (GEC).

GEC is a field that focuses on the automated rectification of various types of errors. These errors include, but are not limited to, spelling errors, grammar errors, collocation errors, and other errors that are closely associated with these aforementioned categories. All of these errors occur within a given sentence. The primary objective of GEC is to furnish invaluable and pertinent information to second-language learners, thereby facilitating their language acquisition process[3], [4].

GEC has been extensively studied for widely spoken languages like English with many existing systems and datasets. One of the most widely employed datasets in the field is the LANG-8 Learner Corpus[3]. This corpus spans 80 different languages and established through a collaborative effort of language learners providing corrections to each other's written texts. However, there has been only a limited research and resource on error correction of other languages[5].

Over the past decade, the field of GEC has witnessed notable progress due to a series of collaborative efforts. Early approaches relied on rule-based methods, followed by the adoption of statistical classifiers. More recently, the field has shifted towards Statistical Machine Translation (SMT) techniques, further advancing GEC capabilities. Currently, Deep Learning (DL) models[6] are the most advanced methodology used in GEC[1]. These progresses are primarily because of its capacity to enhance both the grammaticality and readability of texts generated by users and machines[3].

The development of GEC systems has been significantly enhanced by the availability of extensive parallel data corpuses[1], [7]. These corpuses include millions of sentences featuring both grammatically correct and incorrect structures. This high-quality corpus is crucial for the advancement and improvement of GEC systems. Furthermore, the progress in large pre-trained models[8], [9] have led to remarkable achievements in academic GEC systems[3], [10]. These systems have demonstrated promising performance on benchmarks and have become essential components of modern writing assistance applications, such as Google Docs, Grammarly, and Effidit[11].

1.2 Motivation of the Research

Effective written communication is highly important in modern society. It allows the exchange of ideas, perspectives, and responses, thereby enabling meaningful connections between individuals. In today's digital landscape, many individuals struggle to maintain proper syntax and grammar. This is particularly prevalent on social media platforms like Facebook, Twitter, Telegram, and Gmail, even in their native language. Users interacting on these platforms often unintentionally introduce spelling and grammatical errors into their text. Also, with the increasing prevalence of machine language generation, there is a growing need to post-process machine-generated text in Afaan Oromo. This encompasses various tasks such as conditioned text generation, machine translation, speech-to-text conversion, and more. Despite the recent surge in DL applications for GEC in numerous languages, Afaan Oromo remains largely under-resourced and unexplored in this domain. This research aims to bridge this gap by investigating the potential of DL models for improving the accuracy and efficiency of GEC in Afaan Oromo.

1.3 Statement of the Problem

Grammatical errors can significantly impair communication clarity and effectiveness. In[1], it is stated that errors can lead to misinterpretations and confusion. Ultimately, it results in decreased reader engagement and hinders the intended message from being conveyed effectively. This problem is particularly acute for under-resourced languages like Afaan Oromo, where limited resources and language-specific challenges often hinder the development of robust GEC solutions.

Traditional grammar checker tools such as rule-based[12] and statistical approaches[12], [13], [14] offer some benefits in terms of interpretability. However, they have limitations when dealing with languages such as Afaan Oromo that lack sufficient resources. The approaches are labor-intensive to develop and maintain as they require manual effort to define and encode language-specific rules. In addition, they also struggle with complex errors and subtle language nuances.

To this end, the rise of DL models has been found to demonstrate their potential for achieving superior performance in GEC compared to other methods[6], [8], [15], [16], [17]. However, existing DL models often suffer from data scarcity. These models also lack adaptation to the unique linguistic features of under-resourced languages, as pointed out by[5]. This results in models that struggle to generalize effectively to resource constrained languages such as Afaan Oromo.

To tackle these, this thesis focuses on leveraging DL techniques to develop advanced language processing models tailored for Afaan Oromo GEC. The purpose is to enhance grammatical accuracy and quality in both machine-generated and human-authored text. This research endeavors to significantly elevate the grammatical correctness and overall linguistic quality of Afaan Oromo text. This will facilitate effective communication and content creation in the language.

1.3.1 Research Questions

The following research questions are addressed in this work:

RQ1: How can a high-quality parallel dataset of Afaan Oromo sentences be prepared to develop GEC models for this under-resourced language?

RQ2: How can existing DL models be adapted for Afaan Oromo GEC, considering its unique features and data scarcity?

RQ3: Which DL architectures offer the best balance of fluency and overall performance for Afaan Oromo GEC?

1.4 Objective of the Thesis

1.4.1 General Objective

The General objective of this thesis is to develop DL models for effective and robust Afaan Oromo GEC.

1.4.2 Specific Objectives

To meet the above general objective, the following specific objectives are set:

1. To review literatures and related current work in the area of GEC.
2. To collect and process a large-scale Afaan Oromo dataset for training, validating and testing models.
3. To develop a DL models for Afaan Oromo GEC.
4. To evaluate the performance of the developed GEC models on the Afaan Oromo language dataset.
5. To investigate and compare the performance of different DL models for Afaan Oromo GEC.
6. To compare the performance of the developed models against existing Grammar Checker for Afaan Oromo.

1.5 Scope of the Thesis

This research is dedicated to the development and assessment of DL models tailored for Afaan Oromo GEC. The study will explore different DL models that incorporate transfer learning methods to harness insights from pre-existing models. These models will undergo evaluation using Afaan Oromo dataset created specifically for this research. The research focus on the limited grammatical errors in Afaan Oromo including subject-adjective agreement errors, subject-verb agreement errors, word order errors, adverb-verb agreement errors, and subject and adverb agreement errors.

1.6 Significance of the Study

Grammatical errors not only hinder communication clarity and accuracy but also negatively impact the perception of text quality and professionalism. GEC tools play a crucial role in enhancing written communication by automatically identifying and correcting grammatical errors. For languages like Afaan Oromo, where accurate and efficient GEC tools are scarce, the need for such technology becomes even more paramount.

This research has the potential to make significant impact in several areas by:

1. Facilitating clearer and more effective communication for both native speakers and learners.

2. Enhancing the quality of machine-generated text through effective post-processing techniques.
3. Supporting educational initiatives, promoting literacy and language learning for Afaan Oromo speakers.
4. Contributing to the development of GEC models for under-resourced languages.
5. Enabling wider access to information and resources in the Afaan Oromo language.

1.7 Contributions of the Study

This research makes several key contributions to the field of GEC including:

1. The study designed a DL-based models to automatically detect and fix grammatical errors in Afaan Oromo text. This is one of the first comprehensive GEC systems developed for Afaan Oromo.
2. The study investigates transformer models for GEC, exploring transfer learning. Different layers are frozen to identify the best architecture. This research contributes to understanding transfer learning's effectiveness.
3. As part of the research, the study constructed a high-quality parallel dataset of correct and incorrect Afaan Oromo sentences. This also contributes a valuable resource for developing Multilingual GEC models.
4. The study presents the improved performance of the DL models compared to the previous work on the hybrid grammar checker for Afaan Oromo.

1.8 Thesis Outline

The remainder of the thesis is organized as follows. Chapter 2 presents a literature review on GEC, providing an overview of the Afaan Oromo grammar and common error types, as well as a review of related works on GEC. Chapter 3 then introduces the proposed DL-based GEC system developed for Afaan Oromo, describing its components. In Chapter 4, the experimental setup, evaluation, and analysis of the proposed Afaan Oromo GEC system's performance are detailed, and its strengths, limitations, and potential improvements discussed. Finally, Chapter 5 concludes the thesis by outlining promising directions for future Afaan Oromo GEC research.

CHAPTER TWO

LITERATURE REVIEW

In this chapter an in-depth examination of existing research and methodologies in the field of GEC will be explored. The concept of grammatical errors will first be explored, followed by the process of GEC, with various techniques and approaches used to identify and rectify errors in a text. The chapter will then focus on an overview of the Afaan Oromo language and its unique features. Finally, the related works in the field of GEC will be reviewed, examining the existing research, methodologies, and tools that have been developed.

2.1 Grammatical Error

Every natural language has its unique grammar, which encompasses the study of morphology, and syntax[18]. Morphology involves the rules that govern word structure and construction, while syntax refers to the rules that govern word sequence and sentence structure.

Errors found in written text can be grouped into different categories, including spelling errors, grammar errors, style errors, and semantic errors[2]. When writing in Afaan Oromo, various types of grammatical errors can occur, such as subject-verb agreement, word order, adjective-noun agreement, adverb-verb agreement, and other errors[2], [18], [19].

Defining a grammatical error can be surprisingly difficult due to its complexity. Some spelling mistakes show a similar occurrence rate among both native and non-native writers and do not have any significant grammatical consequences[1].

2.2 GEC

GEC is a crucial task in Natural Language processing (NLP) that aims to automatically detect and correct grammatical, orthographic, and semantic errors in written text. The core objective is to transform ill-formed or erroneous input into grammatically correct and coherent output. GEC systems leverage large annotated datasets containing examples of errors and their corresponding corrections. This allows them to learn the patterns and transformations needed to fix a variety of language mistakes. By drawing

upon these training examples, GEC models are able to analyze the input text, identify problem areas, and generate appropriate corrections. The development of GEC technology holds promise not only for improving academic writing, but also for enhancing written communication across a wide range of real-world settings beyond formal education.

2.2.1 GEC Approaches

Notable progress has been made in this area over the past decade, particularly in the late 2010s, when near human-level GEC systems became available[20], [21]. This progress has been driven by the increasing popularity of machine learning (ML) and DL approaches[22]. The field has seen the development of various approaches, including SMT-based, Neural Machine Translation (NMT) based, classification-based, and language models (LM) based methods.

In the past, ML classifiers were widely used for GEC tasks. This was primarily because some of the most frequent error types made by English language learners, such as mistakes with articles and prepositions, have a limited set of possible corrections[23]. This made these types of errors well-suited to be addressed using multiclass classification models, which were a popular approach for GEC at the time.

SMT approaches to GEC error correction have the potential to address and fix all types of language errors simultaneously, without the need for extensive feature engineering or reliance on expert linguistic knowledge[23]. This is an advantage over classifier-based approaches, which are more limited in the range of errors they can handle. Although SMT was originally developed for translating between different languages, it has been effectively applied to the task of GEC as well. This is because GEC can be viewed as a translation problem - converting sentences containing errors into their corrected, error-free versions. The translation-based approach of SMT has proven suitable for addressing this GEC task.

The rise of DL and the impressive results achieved by NMT models on translation and other sequence-to-sequence tasks led to the natural extension of NMT approaches to the task of GEC as well[23]. Compared to the SMT approach, NMT uses a single large NN to model the entire error correction process. This eliminates the need for the extensive feature engineering that was required for GEC-specific SMT systems.

Language model-based GEC leverages techniques using n-gram models or Transformer-based language models. This language modeling approach was common before machine translation became the dominant GEC paradigm. However, it has seen a resurgence recently, particularly in low-resource or unsupervised GEC settings, due to the effectiveness of large pre-trained Transformer-based language models[23]. Recent advancements have enabled these Transformer-based language models to better capture syntactic information. This has made them capable of functioning as GEC systems even when only limited training data is available. These language model-based GEC systems can become even more effective when supplemented with a small amount of parallel data for fine-tuning.

Ensembling is a common technique used in machine learning to combine the predictions of multiple models that have been trained individually. Ensembles often generate more accurate predictions than any of the individual models that are combined. In the context of GEC, ensembling refers to averaging the probability distributions output by multiple separately-trained GEC models when predicting the next token. Ensembling can also be used to average the edit tags predicted by multiple GEC models in an edit-based approach[23].

Additionally, performance boosting techniques and data augmentation methods have been applied to enhance GEC systems[1] . The integration of NMT approaches has been particularly influential, with many GEC systems based on NMT models[21], [24]. This shift towards DL has not only improved the accuracy of GEC systems but also opened up new avenues for research and development in the field[25].

2.2.2 GEC in Low-Resource Scenarios

The recent achievements in GEC systems are heavily reliant on the presence of a substantial volume, measured in the magnitude of millions, of parallel data. This parallel data encompasses both grammatically accurate and inaccurate sentences[7]. High-quality and extensive data plays a vital role in the training of efficient models for GEC.

GEC in low-resource scenarios is an emerging area of research. However, the languages with limited resources poses a significant challenge. This is due to the scarcity of high-quality data and computational resources. These scenarios often involve languages with limited resources, such as minority languages, or languages with small corpora.

In recent research, approaches have been explored to tackle the challenge of GEC in low-resource settings. Despite the scarcity of data, researchers have investigated various techniques and methodologies to address the needs of languages with limited resources. These efforts have resulted in innovative approaches being developed to achieve effective GEC in such scenarios.

One innovative approach proposed by[16] involves using artificially generated data through error injection to pre-train neural GEC models. In[5], a new dataset AKCES-GEC was introduced for Czech GEC. The study showed that using a synthetic parallel corpus, pre-trained models were able to achieve improved results on Czech, German, and Russian language datasets. Additionally, a study highlighted the potential of transfer learning techniques to leverage data from high-resource languages and improve error correction in low-resource languages[26], [27].

These highlight the potential of transfer learning, transformer models, and synthetic data creation in addressing GEC in resource-constrained scenarios. Through the application of these approaches, significant improvements in error correction accuracy have been achieved for underrepresented languages.

2.3 An Overview of Afaan Oromo Language

Afaan Oromo is one of the major indigenous African languages that is extensively spoken and utilized in various parts of Ethiopia and some neighboring countries[28], [29]. This expansive geographical distribution has resulted in the emergence of several dialects. In these diverse regions, Afaan Oromo shows variations based on the geographical areas where it is spoken. These variations can be observed in pronunciation, vocabulary, and grammar, reflecting the linguistic adaptations of the Oromo people to their specific environments[30]. Despite these differences, the language maintains a high degree of mutual intelligibility, enabling speakers from different regions to communicate effectively.

2.3.1 Afaan Oromo Writing System

Since 1991, the Afaan Oromo language has been using a Latin alphabet-based writing system called *Qubee*, as its official script[2], [31], [32]. The Afaan Oromo language exhibits distinct sounds produced by its vowels and consonants. It consists of a total of thirty-three (33) letters, with a unique combination of vowels and consonants. Among

these letters, seven are compound consonants: CH, DH, NY, PH, SH, TS, and ZH, while the remaining are single consonants and vowels - A, E, I, O, and U. A group of consonant letters is referred to as "*Qubee Dachaa*" in the language. The vowel letters are called "*Qubee Dubbachiistuu*" and single consonants are called "*Qubee Dubbifamaa*".

In Afaan Oromo, there are five short and five long vowels that can produce distinct sounds independently. The long vowels are derived by doubling the duration of the corresponding short vowels. Additionally, single-consonant character like "m" in "*madda*" are allowed to undergo germination, which represents the process of doubling a letter to modify meaning. However, digraph symbols, which consist of two characters, are not permitted to undergo this process. For instance, the word "*haadha*" meaning "mother" cannot be transformed into "*haadhdha*" by applying germination to its digraph symbol[31]. These variations in vowel length and consonant doubling contribute to nuanced differences in meaning within Afaan Oromo[31].

2.3.2 Afaan Oromo Word Class and Their Morphology

There are different word classes in Afaan Oromo, which consist of noun (*Maqaa*), pronoun (*Ba-maqaa*), adjective (*Ibsa-maqaa*), verb (*Xumura*), adverb (*Ibsa-xumuraa*), conjunction (*Walqabsiistota*), preposition (*Dur-duubee*), and prepositional phrase. These classes exhibit both derivational and inflectional morphology, contributing to the richness and complexity of the language's structure[19].

a. Noun (Maqaa)

According to[32], in Afaan Oromo, the majority of sentences start with a noun that is capitalized and acts as the subject. This noun is then accompanied by subject markers (-ni or -n). The subject noun may be followed by the direct object and indirect object, although this is optional. In Afaan Oromo, words with suffixes like -eenya, -ina, or -umma are nouns.

b. Pronoun (Ba-maqaa)

In Afaan Oromo, pronouns serve the same function as nouns, and they are categorized based on their role and meaning in sentences. These categories include reciprocal, reflective, demonstrative, personal, and possessive pronouns[2]. In Afaan Oromo, the subject pronoun comes at the start of the sentence, while the object pronoun comes

before the verb. This is different from English, where the object pronoun comes after the verb[33].

c. Adjectives (Ibsa-maqaa)

Adjectives are words that describe or modify nouns. They can identify or provide additional information about another person or thing mentioned in the sentence. Typically, adjectives in English precede the noun or pronoun they modify. In Afaan Oromo, however, adjectives follow the nouns they modify[34], [35].

d. Verb (Xumura)

The verb conveys an action, an event, or a state of being. It holds significant importance in a sentence as it provides information about the subject to the reader. In Afaan Oromo, verbs are exclusively positioned at the end of sentences. However, it is not accurate to claim that verbs constitute a distinct category in all languages with an open word class[32], [36].

e. Adverb (Ibsa-xumuraa)

Adverbs in Afaan Oromo are utilized to alter verbs and adverbs, often appearing before the verbs they modify. These adverbs can be classified into adverbial time, adverbial place, and adverbial condition categories as noted by[2], [37]. Adverbial time encompasses terms such as *amma* (now), *yoom* (when), *har'a* (today), *kaleessa* (yesterday), *boru* (tomorrow), *iftaan* (after tomorrow), among others. Adverbial places serve to answer the query of 'where' in a sentence. In Afaan Oromo, adverbial place encompasses words like "*as*" (here), "*gubbaa*" (above), "*gidduu*" (middle), "*bira*" (nearby), "jala" (*under*), among others.

f. Adpositions (Dur-duubee)

Adpositions are words that acquire meaning only in conjunction with other words, such as nouns, verbs, pronouns, or adjectives[19]. The distinguishing feature of adpositions is their absence of inflectional or derivational morphology.

g. Conjunctions (Walqabsiistota)

In Afaan Oromo, there are two main types of conjunctions: coordinating and subordinating conjunctions. Coordinating conjunctions are used to join two separate clauses, often when the speaker wants to give equal importance to both sentences[34].

These conjunctions are generally used to combine complete ideas, unlike prepositions which connect nouns to other parts of the sentence. Subordinating conjunctions are used to connect the main clause with a subordinate clause. These conjunctions are always followed by a phrase.

h. Numerals

Words that specify the quantity or order of something are classified under numbers and are positioned after the noun they describe[2], [34]. In Afaan Oromo, numerals are categorized into two types: cardinal and ordinal numbers[36]. Cardinal numerals are used for counting and indicate quantity, while ordinal numerals express the rank or position of items[37]. Ordinal numerals are formed by adding the suffixes *-ffaa* or *-affaa* to the number.

i. Introjections

Afaan Oromo has specific terms to use to convey the feelings and emotions in reaction to unexpected situations. For example, "*Ishoo dhala koo*!" translates to "Wow my child!". Here, "*Ishoo*" serves as an introjection, indicating delight or expressing enjoyment of something[19].

2.3.3 Punctuation marks in Afaan Oromo

The punctuation marks used in both Afaan Oromo and English are identical and serve the same functions, except for the apostrophe. In English, the apostrophe is mainly used to show possession, but in Afaan Oromo, it indicates a glitch sound called 'hudhaa'[2]. For instance, words like "*du'a*" (death), "*har'a*" (today), and "*baay'ee*" (many) include glitch sounds.

2.3.4 Afaan Oromo Grammatical Errors

a. Word order errors

Word ordering errors (WOEs) are instances where the arrangement of words in sentences is incorrect[38]. Different languages have various word orders. These include basic sentence structures, modifiers, and modified word orders such as adjective and noun or adverb and verb arrangements. Languages can exhibit diverse word order patterns, with varying placements of subjects, objects, and verbs[2], [12]. Afaan Oromo sentences typically follow the Subject-Object-Verb (SOV) order as shown in Table 2.1.

Table 2.1: Correct Afaan Oromo word order examples (SOV)

Afaan Oromo sentences	English sentences
Burqaan mana ijaare.	Burka built a house.
Kennaan Leenca ajjeese.	Kenna killed a lion.

In Table 2.1, *Burqaan* (Burka) and *Kennaan* (Kenna) are the subjects of the sentence, *mana* (a house) and *Leenca* (a lion) are the objects, and *ijaare* (built) and *ajjeese* (killed) are the verbs.

A sentence is considered grammatically incorrect if the placement of its words is inappropriate. This encompasses the order of adjective and noun, adverb and verb, verb and sentence end, and other word classes within a sentence. In Afaan Oromo, adverbs should precede the verb they modify, while adjectives should follow the noun they modify, as stated by[2]. This means that the word order in Afaan Oromo is such that adverbs are placed before the verb, and adjectives are placed after the noun they modify as can be seen from Table 2.2.

Table 2.2: Afaan Oromo word order error examples

Incorrect Afaan Oromo sentences	Corrected Afaan Oromo Sentences	English sentences
Beekaan bareedduu mucoo fuudhe.	*Beekaan mucoo bareedduu fuudhe.*	Beka married the beautiful girl.
Isheen hojiishee hojjette sirriitti.	*Isheen hojiishee sirriitti hojjette.*	She worked her job correctly.

As shown in the Table 2.2, on the first incorrect example, the adjective "*bareedduu*" (beautiful) comes before the noun "*mucoo*" (a girl) that it modifies, which is not the standard Afaan Oromo word order. The correct Afaan Oromo order is noun followed by adjective, as shown in the first correct example: "*Beekaan mucoo bareedduu fuudhe*" (Beka married the beautiful girl).

On the second incorrect example, the adverb "sirriitti" (correctly) comes after the verb "hojjette" (worked), whereas the typical Afaan Oromo word order places the adverb before the verb. This is demonstrated in the second correct example: "Isheen hojiishee sirriitti hojjette" (She worked her job correctly).

b. Subject-verb agreement errors

In Afaan Oromo sentences, a subject, which can be a word, noun phrase, pronoun, or equivalent, is typically found at the beginning. The subject is then used by the rest of the sentence to make a statement and agree with the verb, usually indicating an action performed by the verb. The subject of a sentence must agree with the verb in terms of gender, number, and person[39].

Table 2.3: Subject-Verb agreement error in terms of gender

Incorrect Afaan Oromo sentence	Corrected Afaan Oromo Sentences	English sentences
Inni lukkuu bitte.	*Inni Lukkuu bite.*	He bought a chicken.
	Isheen Lukkuu bitte.	She bought a chicken.

In Table 2.3 on the incorrect sentence, there is a disagreement in gender between the subject "*Inni*" (He) and the verb "*bitte*" (bought). To resolve this, either the subject or the verb can be changed, as shown in the corrected sentences.

In the context of numbers, the subject of a sentence must also align with the verb. As shown in the Table 2.4 on the incorrect sentence, "*Dhukkubsattoonni*" (patients) is the subject in plural form, while "*yaalame*" (treated) is the verb in singular form. This mismatch in number between the subject and verb can be corrected by adjusting either the subject or the verb, as shown in the corrected sentences.

Table 2.4: Subject-Verb agreement error in terms of number

Incorrect Afaan Oromo sentence	Corrected Afaan Oromo Sentences	English sentences
Dhukkubsattoonni yaalame.	*Dhukkubsattoonni yaalaman.*	The patients were treated.
	Dhukkubsataan yaalame.	The patient was treated.

In terms of person, the subject of a sentence must also match the verb. In the Table 2.5 on incorrect example, "*Ani*" (I) is the subject in the first person, while "*bitan*" (bought) is in the third person. This mismatch in person between the subject and verb can be corrected by adjusting either the subject or the verb, as shown.

Table 2.5: Subject-Verb agreement error in terms of person

Incorrect Afaan Oromo sentence	Corrected Afaan Oromo Sentences	English sentences
Ani konkolaataa bitan.	*Isaan konkolaataa bitan.*	They bought a car.
	Ani konkolaataa bite.	I bought a car.

c. Subject-adjective agreement errors

In Afaan Oromo, adjectives are words used to describe or modify another person or thing in a sentence[39]. Adjectives in Afaan Oromo must agree with the noun they are describing in terms of both number and gender. They can indicate the number (singular or plural) and gender (feminine, masculine, or neutral) of the noun they are modifying, and therefore, they need to match the noun in number and gender[2], [39].

Table 2.6: Subject-Adjective agreement errors examples

Incorrect Afaan Oromo sentence	Corrected Afaan Oromo Sentences	English sentences
Dubartoota bareedduun arge.	*Dubartii bareedduun arge.*	I saw a beautiful woman.
	Dubartoota babbareedoon arge.	I saw beautiful women.

In Table 2.6 on incorrect sentence, "*Dubartoota*" as the subject indicates a plural subject, while "*bareedduu*" as the adjective indicates a singular subject, resulting in an agreement error between the subject and the adjective. To rectify this error, either the subject or the adjective can be modified, as shown in the corrected sentences.

d. Adverb-verb agreement errors

In Afaan Oromo, adverbs typically modify the first verb that follows them, as noted by[12], and are generally the words that alter any linguistic aspect other than a noun. Adverbs can modify clauses, phrases, verbs, adjectives, and other adverbs, as emphasized by[39]. A common grammar mistake in Afaan Oromo, especially for beginners, is the conflict between the time adverb and the verb tense[2].

Table 2.7: Adverb-Verb agreement error examples

Incorrect Afaan Oromo sentence	Corrected Afaan Oromo Sentences	English sentences
Inni kaleessa dhufa.	*Inni kaleessa dhufe.*	He came yesterday.
	Inni boru dhufa.	He will come tomorrow.

In Table 2.7 on incorrect sentence, "*kaleessa*" (yesterday) is a time adverb indicating the past, but "*dhufa*" (will come) is a verb describing a future action. This creates a disagreement between the adverb and the verb. To correct this, either the adverb or the verb can be changed as shown on the corrected sentences.

e. Subject and adverb agreement error

Discrepancies between subjects and adverbs occur in Afaan Oromo when the subject of a sentence does not match the adverb, especially in the first person singular and when dealing with time adverbs in sentences[19].

Table 2.8: Subject-Adverb agreement error examples

Incorrect Afaan Oromo sentence	Corrected Afaan Oromo Sentences	English sentences
Ani kaleessa dhufe.	*Ani kaleessan dhufe.*	I came yesterday.
Ani bori dhufa.	*Ani borin dhsufa.*	I will come tomorrow.

In Table 2.8 on example sentences, the subject "*Ani*" and the time adverbs "*kaleessa*" or "*bori*" do not agree with each other. To correct this, *"-en, -un,- an, -in,-n*" letters must be added to the time adverbs as in corrected version.

2.4 DL Models for GEC

DL Models for GEC have emerged as significant advancements in the field of NLP, particularly in the context of enhancing GEC systems. This introduction has moved the field away from traditional rule-based and statistical methods towards more sophisticated, AI-driven solutions. This has led to significant improvements in text analysis, error detection, and correction mechanisms.

The use of DL models for GEC has brought about a revolution in the field, addressing limitations present in traditional GEC methods. These models have proven to be powerful tools, providing enhanced efficiency and accuracy in error correction. With

their ability to automatically learn intricate patterns and relationships in language data, DL has paved the way for the development of more effective and robust GEC systems. This has marked a significant advancement in the pursuit of efficient and accurate error correction[40]. This transition to DL models has been made possible by advancements in parallel computing and the widespread adoption of Neural Networks (NN), resulting in significant improvements in GEC techniques since the 1980s[40].

The development of Pre-trained Language Models (PLMs) such as BART[41], BERT[8] and T5[42] has made a significant contribution to the field of GEC. These highly potent models, which have been trained on extensive collections of text, have the capacity to capture intricate linguistic representations and patterns. This enables them to effectively handle a wide range of grammatical structures and error types with high precision.

2.4.1 Long Short-Term Memory (LSTM)

LSTM is a form of Recurrent Neural Networks (RNN) designed to address the vanishing gradient problem and capture extensive dependencies in sequential data. By incorporating memory cells and gating mechanisms, LSTM effectively maintains and selectively modifies information over time. It processes sequential data in a unidirectional manner, considering the preceding context of the input sequence for prediction.

The core architecture of an LSTM consists of several essential components: the input gate, forget gate, cell state, and output gate. These components work together to enable LSTMs to process sequential data and maintain long-term dependencies, making them well-suited for tasks like GEC[43], [44]. The input gate uses a sigmoid activation function to determine the relevance and amount of new information to be added to the cell state, while the forget gate, also employing a sigmoid activation function, controls what information should be forgotten from the cell state. Together, the input gate, and forget gate regulate the flow of information within the LSTM cell.

The cell state serves as a memory that carries information from the past, passing it through the LSTM from the input to the output. It is updated and modified based on the input gate and forget gate, allowing the LSTM to retain important information and discard irrelevant details. Finally, the output gate determines the output based on the current cell state and input, enabling the LSTM cell to produce the final output based

on the processed information. Through these interconnected components, LSTMs can effectively process sequential data, capture long-term dependencies, and generate accurate outputs for various tasks, including GEC.

In a patent titled "Deep context-based grammatical error correction using artificial neural networks," LSTM models are employed as part of a system for grammatical error detection. The system identifies target words in a sentence based on grammatical error types. It then estimates a classification of the target word with respect to the corresponding grammatical error type using an ANN model trained for the grammatical error type. The model includes two RNN configured to output a context vector of the target word based on at least one word before and after the target word in the sentence. This context vector is then used by a feedforward NN to output a classification value of the target word with respect to the grammatical error type. This approach leverages the LSTM's ability to capture the context of a word within a sentence, enhancing the accuracy of grammatical error detection and correction[45].

2.4.2 Transformer Models

The Transformer model, introduced by[46] represents a significant advancement in the field of NLP and DL. It has become the foundation for many state-of-the-art models, including BERT, BART, T5, and others, due to its innovative approach to processing sequential data.

The Transformer model is based on the concept of self-attention mechanisms[46]. These mechanisms allow it to weigh the importance of different words in a sentence relative to each other. This is a departure from previous models that relied on RNNs or CNNs to process sequential data. The Transformer model processes all words in a sentence simultaneously, making it more efficient and effective for understanding the context of words in a sentence.

The architecture of the Transformer model consists of two main components:

1. Encoder: This processes the input sequence and generates a context vector for each word in the sequence. This is done through a series of self-attention layers and feed-forward NNs.

2. Decoder: This generates the output sequence, one word at a time, based on the context vectors provided by the encoder. It also uses self-attention mechanisms to weigh the importance of different words in the output sequence.

a. T5

T5 is a transformer-based model that treats every NLP task as a text generation problem[47]. T5 is structured as an encoder-decoder model, where the input text undergoes encoding to produce a context vector, which is then decoded to generate the output text. This structure empowers T5 to handle tasks that necessitate comprehending the contextual information within the input text and generating appropriate responses or corrections[47]. To achieve this capability, T5 undergoes a two-step training process.

Firstly, T5 is pre-trained on a vast corpus of text using the Causal Language Modeling (CLM) objective. This pre-training process allows the model to acquire an understanding of the statistical properties of language, enabling it to generate text that is both coherent and contextually relevant. Secondly, T5 is fine-tuned on task-specific datasets, such as GEC. During this fine-tuning stage, the model is trained on a dataset comprising sentences with grammatical errors and their corresponding corrected versions. Through this training, T5 learns to identify and rectify errors based on the patterns it has learned during pre-training, as well as the specific examples encountered during fine-tuning[27]. By combining pre-training and fine-tuning, T5 can effectively address grammatical errors and deliver accurate corrections.

b. BART

BART is an innovative NN architecture that combines bidirectional and auto-regressive approaches, resulting in enhanced performance for text generation tasks. Developed by[9], BART functions as a denoising autoencoder and utilizes masked language modeling for Natural Language Generation and Translation. It merges elements from BERT and Generative Pre-trained Transformer (GPT) models, making it a versatile tool for tasks such as text auto-completion, summarization, and generation.

In terms of pretraining, BART[9], [41] predicts an original sequence given a masked and shuffled sequence using a Transformer. It is pretrained on large monolingual corpora that include various domains such as news, books, stories, and web-text.

The denoising autoencoder aspect of BART involves training on corrupted or noisy sentences. The model learns to predict missing or corrupted tokens, resulting in clean

and semantically coherent outputs. BART's strength lies in its ability to handle diverse inputs of varying lengths and formats, making it a powerful model for text generation tasks across different domains. With approximately 140 million parameters, it outperforms models like BERT and GPT-1 due to its combined architecture. This makes it proficient in tasks like GEC, language translation, question-answering, text summarization, and paraphrasing.

c. BERT

BERT, introduced by Google in 2018, is a transformer-based model that revolutionizes NLP by employing bidirectional processing. Unlike traditional models, BERT considers the context of words from both directions, enabling a comprehensive understanding of word meanings. This approach is crucial for accurate grammatical error identification and correction[48].

The core of BERT's architecture lies in transformer blocks, which facilitate parallel text processing, leading to significant efficiency gains over RNNs and CNNs. Through a self-attention mechanism, BERT assigns weights to words in a sentence, prioritizing relevant parts for context comprehension. By processing text bidirectionally, BERT analyzes a word's context from both preceding and succeeding directions, enhancing its ability to grasp word meanings within the entire context[8].

BERT undergoes pre-training on a large text corpus with a focus on two main objectives: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). In MLM, BERT predicts masked words using contextual cues from unmasked words, while NSP involves predicting if two sentences are consecutive. Training on these objectives equips BERT with a strong understanding of language statistics, enhancing its contextual comprehension abilities.

BERT operates by leveraging its understanding of word context, which it acquires through pre-training. The input text is initially encoded into a sequence of vectors using transformer blocks. During this encoding process, the self-attention mechanism is applied to assign weights to individual words, considering their importance within the context of the entire sentence. Notably, BERT processes the encoded input bidirectionally, examining the context of each word from both preceding and succeeding directions. This comprehensive analysis enables BERT to grasp the

meaning of a word based on its complete surroundings, facilitating accurate identification and correction of grammatical errors.

Unlike T5, BERT does not generate text directly. Instead, it provides representations of the input text that can be employed for various NLP tasks, including GEC. These representations capture the contextual information of each word within the sentence. Other models or algorithms can utilize these representations to identify and rectify errors in the text[8].

2.5 Related Works

A wide range of studies have been proposed and carried out regarding GEC for a number of languages, such as English, Arabic, Czech, German, and Russian[1], [3], [17], [49], [50]. Research has been suggested and conducted for Afaan Oromo as well[2], [12], [19] using various methodologies. Numerous GEC use different approaches, each with a unique methodology. The Transformer architecture was used in the majority of recent neural GEC models, and the results were on par with or better than those of convolutional and recurrent architectures[51]. Prior studies on low-resource languages concentrated on GEC classifier methods based on rules, GEC derived from SMT, and traditional NMT methods[49]. This section will offer a thorough analysis of the various GECs currently in use.

The paper[41] delves into utilizing BART as a generic pretrained encoder-decoder model for GEC tasks. The authors demonstrate that both monolingual and multilingual BART models exhibit high performance in GEC, comparable to previous baselines in English GEC. This approach eliminates the need for additional pretraining, offering a more efficient and effective method compared to previous techniques that relied on extensive pretraining with pseudo data. The impressive performance of BART models on GEC benchmarks solidifies their position as robust baselines for future GEC research.

The authors of paper[51] examined a neural GEC system. This system integrates pre-training techniques involving realistic errors, sequential transfer learning, and context-sensitive spelling verification utilizing a neural language model. The papers employ a methodology that involves generating erroneous versions of large unannotated corpora using a realistic noising function. These parallel corpora are then utilized to pre-train

Transformer models. Subsequently, through sequential transfer learning, these models are adapted to the domain and style of the test set. Additionally, a context-aware neural spellchecker is integrated into the system to enhance its performance in GEC tasks. The study reports that their system achieves competitive results in both restricted and low-resource tracks in the ACL 2019 BEA Shared Task. The paper utilizes various datasets such as WI-LOCNESS and WikiText-103 for training and to evaluate the effectiveness of their neural GEC system. A potential limitation of the paper is the need for additional research to assess the system's generalizability across languages and text genres, as well as its robustness in handling diverse grammatical errors. Further investigation into these aspects would be valuable for future research.

In[52] the authors explored the capabilities of GPT-3.5 models in the task of GEC. The author notes that while GPT-3 models have shown high performance in various Natural Language Tasks, there is a lack of detailed analysis of their performance in GEC. To address this gap, the authors conduct experiments using a GPT-3.5 model on major GEC benchmarks. They compare the performance of different prompts in both zero-shot and few-shot settings and analyze the outputs obtained with different prompt formats, highlighting any intriguing or problematic results. The authors specifically focus on the BEA-2019 and JFLEG datasets, evaluating the performance of their best prompt. The findings of the experiments reveal that the GPT models can perform well in a sentence-level revision setting. The authors also compare the corrections made by the GPT models to source sentences, human reference sentences, and a baseline GEC system, observing differences in editing strategies and how these corrections are scored by human raters. Human evaluation experiments are conducted to assess the quality of the models' corrections. A potential limitation of the paper is that it does not include an analysis of GPT-4 in the context of GEC.

The paper[53] proposes a language-independent approach to enhance the efficiency of GEC by splitting the task into two subtasks: Erroneous Span Detection (ESD) and Erroneous Span Correction (ESC). ESD identifies grammatically incorrect text spans using a sequence tagging model. ESC utilizes a seq2seq model to correct the identified erroneous spans in the input sentence. This methodology has the potential to enhance efficiency by directing the correction process towards the text portions with higher likelihood of being erroneous. The datasets employed in this study encompassed FCE, Lang-8, NUCLE, and W&I+LOCNESS, which served as the training data. To identify

the most effective models, the CoNLL-2013 test set was adopted as the development set. The assessment phase involved the usage of well-established GEC benchmark datasets, including CoNLL-2014 and the BEA-2019 test set. The evaluation process for CoNLL-2014 employed the official evaluation script M^2 Scorer, while the BEA-2019 test set was evaluated using ERRANT. Experiments conducted in the study demonstrate that this approach performs comparably to traditional seq2seq methods in English and Chinese GEC benchmarks, achieving similar results with a reduced inference time of less than 50%. However, the paper does not provide a detailed analysis of the error cases of the proposed approach.

In[49] paper, the authors introduce an Indonesian GEC framework that utilizes multi-head attention networks, Transformer, with a copying mechanism to address the challenge of unfamiliar and specialized words commonly found in the source. The paper presents a multi-level, semi-supervised confusion approach for producing synthetic parallel training data from an out-of-domain corpus. It creates a significant quantity of synthetic data to tackle the issue of insufficient training data. The origin of the synthetic data was the open-source monolingual corpus known as the CC100-Indonesian dataset. The designed model possesses the capability of auto-correction for all forms of errors in Indonesian GEC, with the intention of surpassing existing methods, showcasing its versatility. The experimental evaluation of the proposed model demonstrates its significant performance improvements over previous GEC models. The achieved average F1 score of 0.7194 and BLEU score of 78.13 highlight the effectiveness of the model in GEC tasks. The proposed method struggles with complex errors and offers limited control over error types. It also hasn't been tested on less common languages. This suggests future work on generating harder errors, better error control, and broader evaluations.

In[3] the authors present an innovative approach for training advanced multilingual models for GEC. They outline a two-step strategy for training these models across multiple languages. Initially, they introduce a language-independent technique to generate a substantial volume of synthetic examples, addressing the scarcity of high-quality training data for GEC in diverse languages. The subsequent step involves utilizing large-scale multilingual LM with up to 11B parameters for fine-tuning on language-specific supervised datasets. This method harnesses the capabilities of large-scale models to enhance GEC performance across various languages. The paper

introduces the gT5 model, which is employed to refine the targets of the widely used yet noisy Lang-8 dataset, resulting in the creation of the CLANG-8 dataset. The proposed approach outperforms previous state-of-the-art achievements on GEC benchmarks in English, Czech, German, and Russian. It is noted that the language distribution is heavily skewed, with Japanese and English having the most significant presence, each comprising over a million ungrammatical-grammatical sentence pairs, while only ten languages have more than 10,000 sentence pairs each.

The[25] paper introduces a method to enhance GEC systems by providing explanations for the corrections made. This is achieved by annotating a large dataset, named EXPECT, with evidence words and grammatical error types. The authors propose several baselines and analysis to understand this task. The study demonstrates that the proposed explainable GEC system can assist second-language learners in determining whether to accept a correction suggestion and in understanding the associated grammar rule. The paper demonstrates that incorporating explanations into GEC systems can significantly assist second-language learners in understanding the corrections suggested by the system. One limitation of the work discussed is the lack of thorough investigation into seq2seq architectures for explainable GEC. This limitation highlights the need for further exploration and development in this area to enhance the explainability and effectiveness of GEC systems.

In paper[19] the Afaan Oromo grammar checker proposed utilizes a hybrid approach combining statistical and rule-based methods to address word order and morphological agreement errors in Afaan Oromo sentences. The statistical grammar checker module assesses word order errors, while the rule-based module focuses on morphological agreement errors, leveraging language grammar rules. The experiment employs a POS tag set from[34], with a manually prepared POS tagger corpus for training and testing a Hidden Markov model Viterbi model. Additionally, a tag sequence corpus and 150 manually created rules are utilized for the statistical bigram model and rule-based grammar checker, respectively. The evaluation of the Afaan Oromo grammar checker yielded 82% precision, 76% recall, and 79% F-measure on test sentences, showcasing the effectiveness of the hybrid approach. However, limitations may arise in scalability and generalization, suggesting future work involving a high-quality Afaan Oromo corpus and DL techniques to enhance performance further.

Table 2.9: Summary of Related works

Articles	Contributions	Limitations
[3]	❖ Introduced a language-independent technique to generate a substantial volume of synthetic data. ❖ The paper introduces the gT5. ❖ The authors release the CLANG-8 dataset which is a cleaned version of the LANG- 8 dataset.	❖ CLANG-8 dataset is not available for all languages. ❖ lack of extensive evaluation on low-resource languages. ❖ Lack of Comparative Analysis with Other Models.
[25]	❖ Incorporates explanation into GEC Systems.	❖ Focuses on specific languages or contexts. ❖ The evaluation of the explanations provided by the GEC system is somewhat qualitative.
[19]	❖ Showcases effectiveness of hybrid approach.	❖ Small and low-quality dataset. ❖ Error correction for morphological agreement errors only.
[41]	❖ Reveals that monolingual and multilingual BART models achieve high performance in GEC. ❖ Eliminated need for additional pretraining.	❖ Lack of a comprehensive comparison of different pretrained models.
[49]	❖ It creates a significant quantity of synthetic data. ❖ Significant Performance improvement over previous GEC models.	❖ Struggles with complex errors. ❖ Limited controls over error types.

[51]	❖ Models are adapted to the domain and style of the test set.	❖ Systems generalizability across languages and text genres.
[52]	❖ Reveals that GPT models can perform well in a sentence-level revision setting.	❖ The study employs prompt based method. ❖ Does not include analysis of GPT-4.
[53]	❖ Proposed language-independent approach. ❖ Reduced inference time.	❖ Does not provide a detailed error analysis.
[54]	❖ Proposed a unified strategy for multilingual GEC. ❖ Creates diverse parallel GEC data without language-specific operations.	❖ The quality of the synthetic data may be limited by the NAT model's performance. ❖ Lack of Comparison with Other Approaches.

This chapter presents the advancements made in the field of GEC. It has also emphasized the potential of applying cutting-edge DL techniques to low-resource language scenarios, such as Afaan Oromo. Building upon this foundation, Chapter 3 will delve into the methodological approach adopted in the current study to thoroughly investigate the application of DL models for GEC in the Afaan Oromo language. The research design, data collection procedures, and analytical techniques will be detailed to systematically evaluate the effectiveness of this approach and uncover strategies for overcoming the challenges identified in the existing literature.

CHAPTER THREE

METHOD AND DESIGN

This chapter explains the methods used to conduct the research, including preparing the dataset, pre-processing the data, selecting the model, training the model, and evaluating the model's performance.

3.1 Proposed Solution

The study employs a structured approach to address the challenge of GEC in Afaan Oromo. The research methodology is designed to systematically tackle the research objectives. Initially a thorough review of the existing literature on GEC techniques, particularly in the context of low-resource languages like Afaan Oromo, is conducted. A dataset is then gathered and preprocessed to ensure compatibility with the proposed model. The data preprocessing involves cleaning, tokenizing, and formatting the text to align with the model's input requirements.

A Transformer-based architecture with transfer learning is leveraged for the model development. Transformer models are known for their strong performance in various NLP tasks, including GEC. The study fine-tunes a pre-trained Transformer model on the Afaan Oromo dataset. To fine-tune the model effectively, certain lower parts of the encoder and decoder layers of the Transformer architecture are frozen, while the upper layers are trained on the target dataset. This approach helps the model to retain the learned representations from the pre-trained model's encoder and decoder, while allowing the upper layers to adapt to the specific task and dataset. By freezing the lower layers, the model requires fewer parameters to be trained, which can significantly reduce the overall training time and computational resources needed.

The model is trained and tested using the prepared dataset and specific evaluation metrics. Various fine-tuning techniques and hyperparameter optimization strategies are explored to further enhance the model's performance. The proposed solution is depicted as in Figure 3.1.

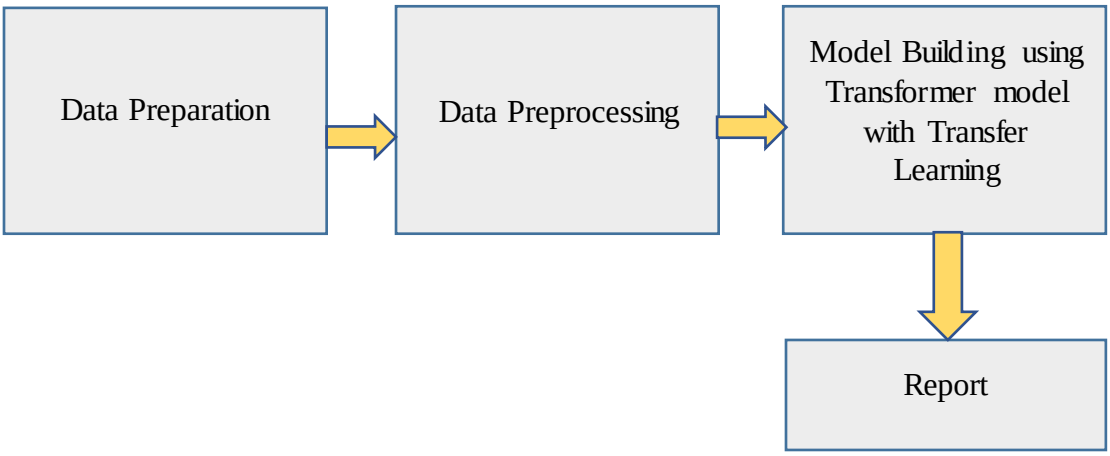


Figure 3.1: Proposed Solution for GEC in Afaan Oromo

3.2 Data Preparation

For this research, a parallel dataset of approximately 10,000 Afaan Oromo sentences is initially manually compiled. The dataset is sourced from various platforms, including:

1. Facebook Pages: Text posts and comments from public pages related to the Afaan Oromo language and culture are gathered.
2. Telegram Channels: Text messages from public channels discussing Afaan Oromo-related topics are collected.
3. Afaan Oromo Textbooks: Sentences are extracted from digitized language textbooks and educational materials.
4. Native Afaan Oromo speakers are involved in creating additional sentences to ensure the dataset's comprehensiveness and quality.

The collected sentences are then paired with their grammatically incorrect versions to form the parallel dataset. To produce these incorrect versions, the following error types are applied:

1. Word Order Errors: The order of words in the sentence is rearranged to create a new, incorrect sequence.
2. Subject-Verb Agreement Errors: The subject or verb is changed to create a mismatch between the two, altering the sentence's grammatical structure.
3. Subject-Adjective Agreement Errors: The subject or adjective is modified to create a disagreement between the two, resulting in a sentence with incorrect subject-adjective agreement.

4. Adverb-Verb Agreement Errors: The adverb or verb is changed to create a mismatch between the two, resulting in an incorrect adverb-verb agreement.
5. Subject and Adverb Agreement Errors: Either the subject or the adverb in the sentence is altered to create a mismatch between the two. This results in a sentence with incorrect subject-adverb agreement, where the adverb no longer properly modifies or describes the subject.

This process involved careful manual review and editing by Afaan Oromo language experts to ensure the accuracy and reliability of the grammatical corrections.

3.3 Data Preprocessing

a. Data Cleaning

The parallel Afaan Oromo sentences undergo a rigorous cleaning process to ensure the quality and consistency of the dataset. This involves several meticulous steps. First, a thorough examination of the dataset reveals and eliminates exact duplicate sentences, thereby eliminating redundancy and maintaining the dataset's uniqueness.

Next, the dataset is carefully checked to ensure that the correct and incorrect versions of the sentences are not identical. Additionally, sentences containing irrelevant characters, symbols, or formatting issues are thoroughly scanned for in the dataset. The irrelevant characters, symbols, or formatting issues are manually removed or corrected to maintain the dataset's integrity and readability.

The Afaan Oromo sentences in the dataset are not all initially correctly spelled. Before the targeted grammatical errors are introduced, each sentence is carefully reviewed. Any spelling mistakes are manually corrected to ensure that the words are all in correct spelling. Once the spelling is corrected, the research is focused solely on introducing the targeted grammatical errors.

Each sentence pair is thoroughly examined to verify the accuracy of the grammatical errors, ensuring the linguistic authenticity of the dataset for the GEC task. The manual review and editing process by Afaan Oromo language experts is dedicated to ensuring the reliability and consistency of the introduced grammatical errors. The accuracy of the spelling corrections is also validated during this process.

This comprehensive cleaning process ensures that the final parallel dataset consists of unique Afaan Oromo sentence pairs. This makes the incorrect versions are genuinely different from the correct versions, and the sentences are free of any technical issues or inconsistencies.

b. Data Transformation

After the dataset cleaning process, several data transformations are performed to prepare the data for the DL models. The first step is tokenization, where the Afaan Oromo sentences are segmented into individual words. Due to the shared character set between Afaan Oromo and English, a well-established English tokenizer is leveraged. The Afaan Oromo sentences are successfully segmented while the linguistic authenticity of the data is maintained. This allows the model to have captured the language's subtleties.

To ensure a consistent input size for the DL model, the sentences are padded or truncated to a fixed length. This step is crucial to maintain the model's architectural requirements and enable efficient processing. The optimal padding and truncation strategies are carefully considered to ensure that the model can effectively learn from the data.

Next, a comprehensive vocabulary of the language is built, and each token is assigned a unique numerical index. This robust representation of the language's lexical diversity enables the model to effectively process and analyze the data. The vocabulary building process is designed to capture the intricate relationships between words and phrases in the Afaan Oromo language, allowing the model to have better understood the language's syntax and semantics.

3.4 Dataset Description

The final dataset consists of 9,500 parallel Afaan Oromo sentences, where each sentence is paired with its grammatically incorrect and corrected versions. The dataset covers a wide range of topics, including news, social media discussions, educational materials, general conversational language, and various aspects of Afaan Oromo culture. The dataset includes various types of grammatical errors, such as Subject-adjective agreement errors, Subject-verb agreement errors, Word order errors, Adverb-

verb agreement errors, and Subject and adverb agreement errors. The overall dataset description is shown in the Table 3.1.

Table 3.1: Overall Dataset Description

Dataset Characteristics	**Value**
Language	Afaan Oromo
Total Number of Sentences	9,500
Training Set	7,600 sentences
Validation Set	950 sentences
Test Set	950 sentences
Average Sentence Length	8 words
Vocabulary Size	11,321 unique tokens

The dataset is divided into training, validation, and test sets, with a split ratio of 80:10:10, respectively as depicted in Figure 3.2. This split ensures that the model can be properly trained, validated, and evaluated during the research process.

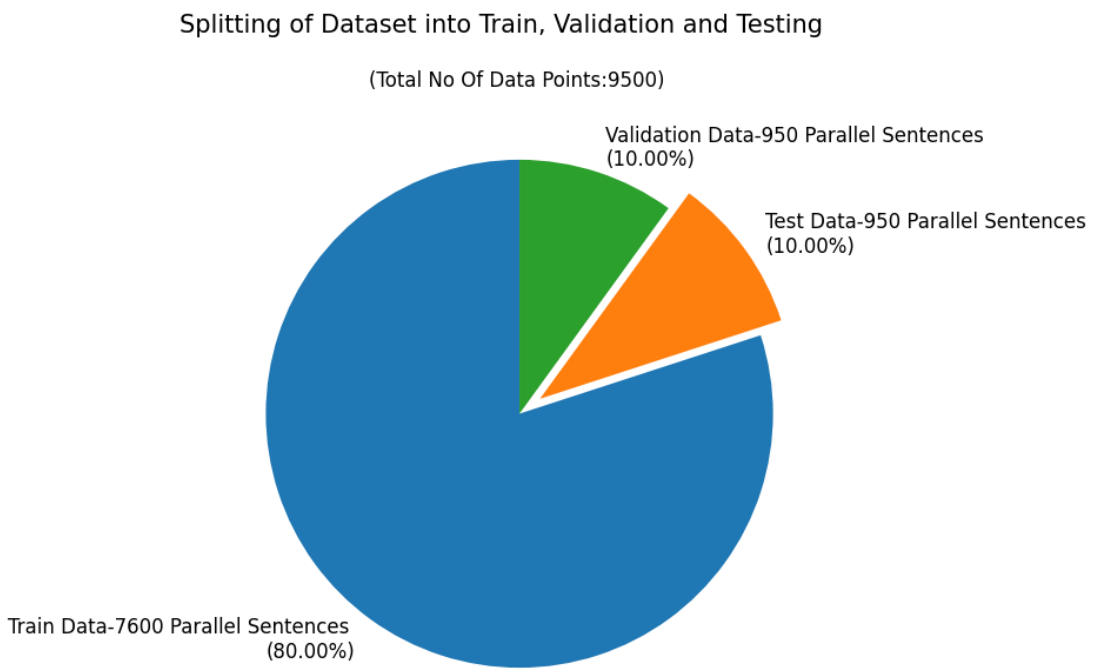


Figure 3.2: Dataset Splitting Strategies

The word count distribution depicted in Figure 3.3 provides valuable insights into the characteristics of the Afaan Oromo GEC Dataset. The long-tailed distribution indicates that the dataset encompasses a broad range of sentence lengths, from relatively brief

sentences to much longer ones. The gradual increase in the slope suggests that there are more sentences with moderate word counts. The steeper part of the curve towards the higher end represents the fewer sentences with very long sentences. This diversity in the dataset is beneficial for training models, as it allows them to learn from a wide range of sentence structures and lengths.

The dataset is stored in a CSV format, with each row containing the incorrect Afaan Oromo sentence and its grammatically corrected version, separated by a comma character.

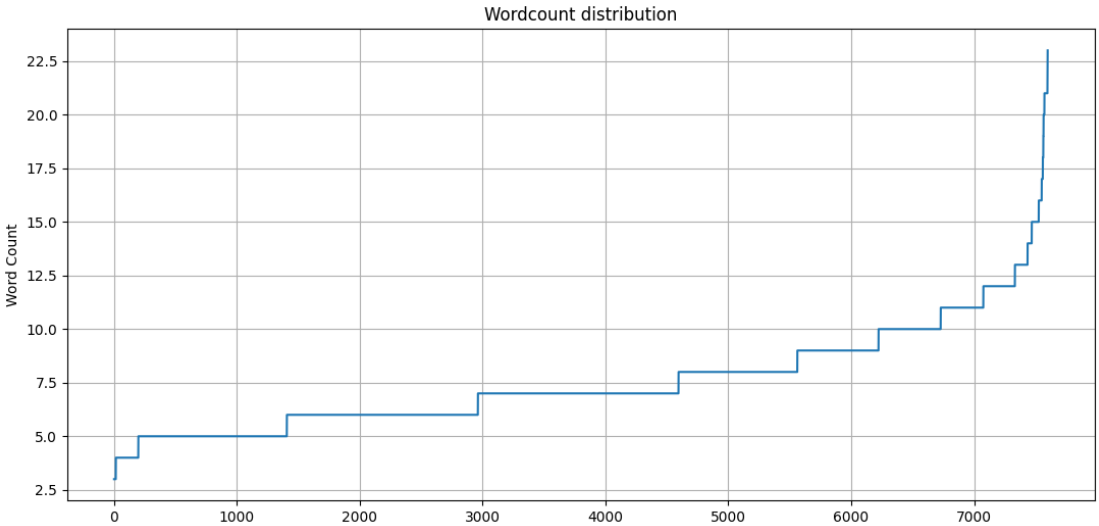


Figure 3.3: Word Count Distribution in the Afaan Oromo GEC Dataset

3.5 Transformer Based Model with Transfer Learning

The pre-trained capabilities of the T5 and BART models are leveraged through a transfer learning approach for fine-tuning the models on the Afaan Oromo GEC task. This transfer learning approach allows the pre-trained models' powerful language generation and understanding capabilities to be benefited from. It also enables the models to learn the nuances of the target language and the task-specific requirements.

This strategy helps to overcome the challenges posed by the limited availability of Afaan Oromo language resources, as the models can leverage the knowledge gained from generic pre-training on larger corpora. It also benefits the time for task-oriented pretraining strategy.

The model development strategy involves freezing various combinations of encoder and decoder layers during training to determine the most effective model architecture.

This allows the contribution of specific components of the model to be assessed and the best configuration for the GEC task in Afaan Oromo to be determined. Based on this the lower-level layers of the encoder and decoder networks are opted to be frozen. These layers are responsible for capturing the general language understanding and representation capabilities learned during the pre-training phase on large-scale corpora.

By freezing the lower-level layers, the models' strong general language understanding abilities are ensured to be maintained. These general language understanding abilities are crucial for effectively handling the Afaan Oromo language. The higher-level layers of the encoder and decoder networks are then trained to adapt the models to the specific task of GEC in Afaan Oromo.

The performance improvements and efficiency gains observed demonstrate the effectiveness of the transfer learning approach. This technique enhances the GEC capabilities for the Afaan Oromo language. Figure 3.4 illustrates the proposed model architecture.

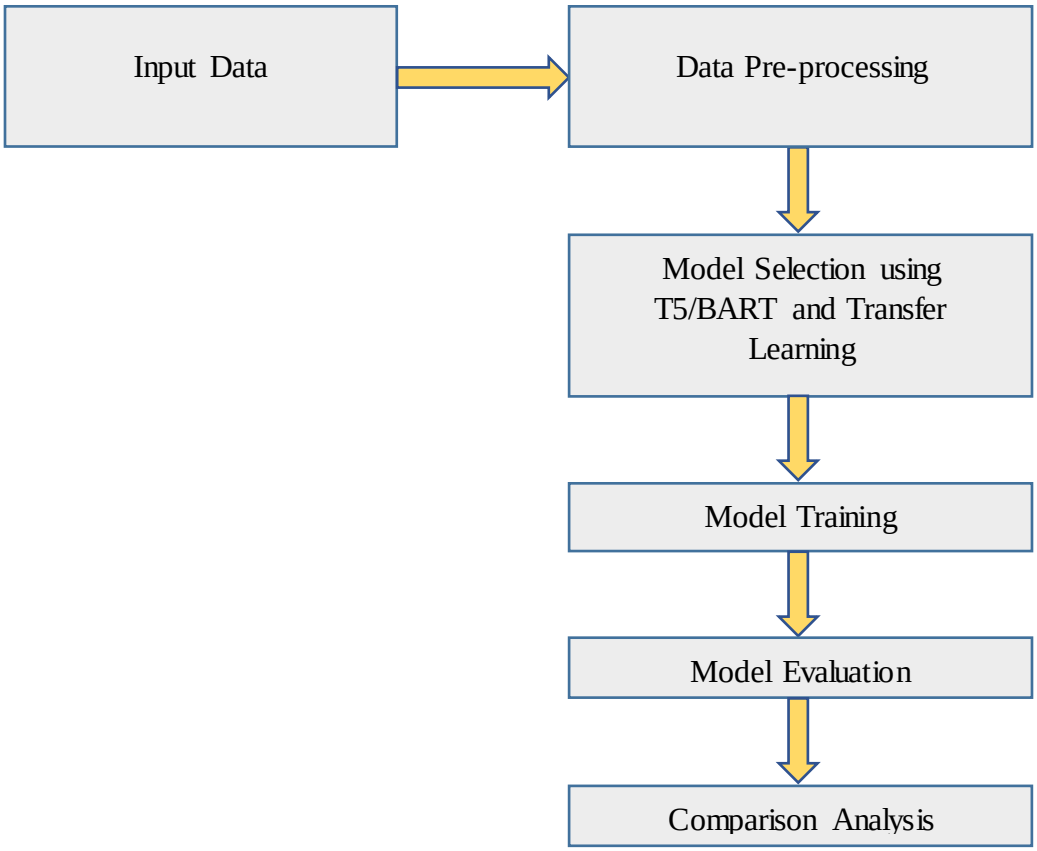


Figure 3.4: Proposed Models

a. GEC Using T5

The approach taken in this work involves fine-tuning the pre-trained T5 model on the Afaan Oromo dataset. By fine-tuning the T5 model, this pre-trained knowledge can be harnessed and adapted to the specific requirements of GEC in the Afaan Oromo language. This allows the model to learn the patterns and rules for correcting grammatical errors in the language. The input to the T5 model is the potentially erroneous sentence, and the output is the corrected version.

The T5-base model, which has 12 encoder layers and 12 decoder layers, is used to capture the complexities of the Afaan Oromo GEC task. This configuration results in the T5-base model having approximately 220 million parameters.

A custom dataset class is created to handle the input and target text for the GEC task. The T5 tokenizer is loaded from the pre-trained 'T5-base' model. The input and target sentences are tokenized and encoded using the T5 tokenizer, with padding and truncation applied. A training pipeline that utilizes the T5 architecture is implemented, where the input is the potentially erroneous sentence, and the output is the corrected version. Optimization techniques, such as a learning rate scheduler, are incorporated to enhance the model's performance during the fine-tuning process.

b. GEC Using BART

In this study, the potential of the BART model as a versatile pre-trained encoder-decoder architecture for the task of GEC in Afaan Oromo is investigated. During the pre-training stage, the BART model is trained to predict the original sequence of text by receiving a masked and shuffled version of the sequence as input. This pre-training approach allows the model to learn a rich representation of language that can be fine-tuned for the specific task of GEC in Afaan Oromo.

Several modifications to the BART model have been made to better suit the unique linguistic features and challenges of the Afaan Oromo language. The BART-large model, with its 12 encoder layers and 12 decoder layers resulting in around 406 million parameters, is employed in this work. The BART-large model has a larger model size and a more extensive pre-training corpus compared to the original BART model.

A specialized dataset class is created to manage the input and target text for the GEC task. This class features suitable tokenization, padding, and truncation methods. A

training pipeline utilizing the BART architecture is established, where the model receives a potentially flawed sentence as input and produces the corrected version as output. To improve the model's performance during fine-tuning, optimization strategies, including a learning rate scheduler, different optimizers and adjustments to batch size, are implemented.

3.6 Evaluation Metrics

The metrics utilized in this research are frequently employed in NLP tasks, which are reference-based GEC metrics. These metrics measure the system-level correlations with humans on a large dataset of human judgments of GEC outputs. They also involve properly conducting statistical significance tests to assess the accuracy of models. These metrics offer quantitative measures that assist in evaluating the model's performance in detecting and rectifying grammatical errors within the language. The following are commonly used performance evaluation metrics along with their respective formulas:

a. General Language Evaluation Understanding (GLEU)

GLEU is an evaluation benchmark designed to measure the performance of language understanding models in a range of NLP tasks. It provides a standardized set of diverse NLP tasks, allowing researchers and practitioners to evaluate and compare the effectiveness of different LM on these tasks. GLEU computes the precision of n-grams in the hypothesis that match part of the reference sentence, similar to the MT metric Bilingual Evaluation Understudy (BLEU). The GEC system's accuracy is determined by comparing a reference sentence with the source sentence. It places greater emphasis on correctly corrected errors, rewarding accurate corrections and uncorrected source text[55]. Additionally, GLEU penalizes n-grams in the hypotheses that match part of the input but not the reference[55], [56].

The computation of GLEU is done as follows. Consider a set of input sentences $S = \{s_1, \ldots, s_n\}$, their corresponding corrected hypotheses $H = \{h_1, \ldots, h_n\}$, and reference sentences, $R = \{r_1, \ldots, r_n\}$.

$$\text{GLEU}\ (\text{S}, \text{H}, \text{R}) = \text{BP}.\exp\left(\frac{1}{N}\sum_{k=1}^{N} \log P_k\right), \tag{2.1}$$

where $BP = \begin{cases} 1, & if\ l_h > l_r \\ \exp(1 - l_r/l_h) & ,\ if\ l_h \leq l_r \end{cases}$ and

$$P_k = \frac{N(H,R) - (N(H,S) - N(H,S,R)}{N\ (H)}$$

Here, *BP* is a brevity penalty that is computed to penalize short hypotheses and P_K is a precision term computed for n-grams of size k (k = 1, 2, …, N and N = 4 for standard GLEU).

b. Max-Match (M^2)

The M^2 metric is a widely used reference-based evaluation metric for assessing the performance of GEC systems[55]. The M^2 metric aims to better capture the quality of GEC system outputs by measuring the maximum overlap between the system's corrections and the reference corrections. It does this by generating the optimal alignment between the system output and reference, and then computing the F-score based on this alignment[56].

The M^2 metric considers a set of input sentences $S = \{s_1, ..., s_n\}$ and the corresponding system-generated hypotheses $H = \{h_1, ..., h_n\}$. For each input sentence s_i, there is a set of gold-standard edits g_i that represent the reference corrections. The set of edits made by the system to transform the input s_i into the hypothesis h_i is denoted as e_i. Precision is calculated by dividing the number of correct edits made by the system by the total number of edits made by the system. Recall is determined by dividing the number of correct edits made by the system by the total number of gold-standard edits. These can be expressed as:

$$\text{Precision} = \frac{\sum_{i=1}^{n} |e_i \cap g_i|}{\sum_{i=1}^{n} |e_i|} \tag{2.2}$$

$$\text{Recall} = \frac{\sum_{i=1}^{n} |e_i \cap g_i|}{\sum_{i=1}^{n} |g_i|} \tag{2.3}$$

$$F_\beta = (1 + \beta^2).\frac{P\ .\ R}{(\beta^2.P) + R} \tag{2.4}$$

In the standard M^2 metric, a β value of 0.5 has been employed since the CoNLL-2014 shared task, which gives double weight to precision compared to recall[56]. This approach places a greater emphasis on penalizing incorrect feedback. This is a crucial

factor in language learning, as it is essential to minimize the risk of providing erroneous guidance to learners. When multiple annotation sets from different annotators are available for a given input sentence s_i, the set that achieves the highest F-measure is selected as g_i.

This chapter outlines the methodology for investigating the use of DL techniques for GEC in the Afaan Oromo language. It describes the data collection, proposed models, and analysis approaches, establishing a foundation to evaluate the models' effectiveness. The next chapter will present the results and discussion of the empirical study, providing a detailed analysis of the performance metrics and insights from the DL-based GEC system. The discussion will also examine the strengths and limitations of the research, guiding future refinements and expansions of this line of inquiry.

CHAPTER 4

RESULT AND DISCUSSION

This chapter presents the results of the proposed models in a systematic and comprehensive manner. A comprehensive set of experiments evaluates the performance of each model and improves the results through various experiments on the proposed approach. The experimental findings briefly summarize, and a comparative analysis conducts between our models and existing system, providing insights into the effectiveness and advantages of our models.

4.1 Experimental Setup

4.1.1 Hardware

The research conducts using the hardware resources provided by the Kaggle environment. The GPU available for free access is Nvidia Tesla P100 GPUs, with a maximum runtime of 12 hours per single session and up to 16GB of RAM, with a maximum of 29GB of total memory. This hardware configuration allows for the efficient training of the DL models used in the study. It also allows for the efficient evaluation of these DL models. The hardware configuration leverages the high-performance capabilities of the Nvidia Tesla P100 GPUs, which accelerate the computationally intensive tasks involved in GEC for the Afaan Oromo language. By taking advantage of the powerful Nvidia Tesla P100 GPUs, the research is able to efficiently train and evaluate the DL models used in the GEC study for the Afaan Oromo language.

4.1.2 Software

The software used for this research includes the Kaggle environment, a popular platform for data science competitions and hosting projects. The Kaggle environment provides a pre-configured setup for data science tasks, including a Jupyter notebook interface and access to various libraries and tools. This makes the Kaggle environment well-suited for rapid prototyping and testing of DL models. The DL libraries used include TensorFlow, PyTorch, and Transformers. Additionally, (Natural Language Tool Kit) NLTK, a popular python library for NLP tasks, is used. The operating system used is Ubuntu, a popular open-source operating system based on the Linux kernel, and

a programming language is Python, a high-level, interpreted programming language widely used for data science and ML tasks.

4.1.3 Hyperparameter

A set of hyperparameters that are commonly used in the literature and that are experimentally determined to be important for the performance of the models is selected. Additionally, the researchers experiment with different optimizers, including AdamW, RMSProp, and SGD. Table 4.1 presents hyperparameters used.

Table 4.1: Hyperparameters used

Hyperparameters	**Values**
Number of epochs	30
Loss function	Cross-entropy
Batch size	8 and 16
Optimizer	AdamW, SGD, and RMSProp
Activation	SoftMax
Learning rate	5e-05

4.2 Experimental Results

In the experiment stage, the capabilities of T5 and BART models are leveraged to evaluate the effectiveness of the proposed approach for Afaan Oromo GEC. These models are trained using different configurations, including batch sizes of 8 and 16, and the number of training epochs is set to 30. Furthermore, three distinct optimizers are explored: AdamW, RMSProp, and SGD. The performance of the models is evaluated using two different metrics, the GLEU and M^2 metrics.

The model development strategy involves freezing various combinations of encoder and decoder layers during training to determine the most effective model architecture. This allows the contribution of specific components of the model to be assessed and the best configuration for the GEC task in Afaan Oromo to be determined. The different experimental cases also aim to investigate the impact of key hyperparameters and optimization algorithms on the model's performance for the Afaan Oromo GEC task.

CASE I (Batch size = 8 and Optimizer = AdawW) represents the baseline configuration. In this case different encoder and decoders of the models are frozen to identify the best

architecture for the task of GEC in Afaan Oromo. CASE II (Batch size = 16 and Optimizer = AdawW) explores the impact of increasing the batch size, and CASE III (Batch size = 8 and Optimizer = RMSProp) and CASE IV (Batch size = 8 and Optimizer = SGD) explore the use of different optimization algorithms. Analyzing the results of these diverse experimental cases will provide valuable insights into the most effective hyperparameter configurations.

CASE I: Batch size = 8 and Optimizer = AdamW

In the baseline configuration, the performance of the BART and T5 models is explored by freezing different numbers of encoder and decoder layers. This approach allows the impact of layer freezing on the overall model performance to be studied. The results of these experiments are shown in Table 4.2 and Table 4.3.

Table 4.2: Model Evaluation Result for CASE I (T5)

Models	Frozen Layers (The lower part)	GLUE	M^2		
			P	R	$F_{0.5}$
T5	2	0.7562	0.7110	0.5960	0.6799
	4	0.7604	0.7130	0.5990	0.6816
	5	0.7241	0.7644	0.6239	0.7418
	6	**0.7679**	**0.8158**	**0.7489**	**0.8021**
	7	0.7554	0.7649	0.6745	0.7413
	8	0.7630	0.7140	0.6001	0.6806
	10	0.7504	0.7120	0.5960	0.6786

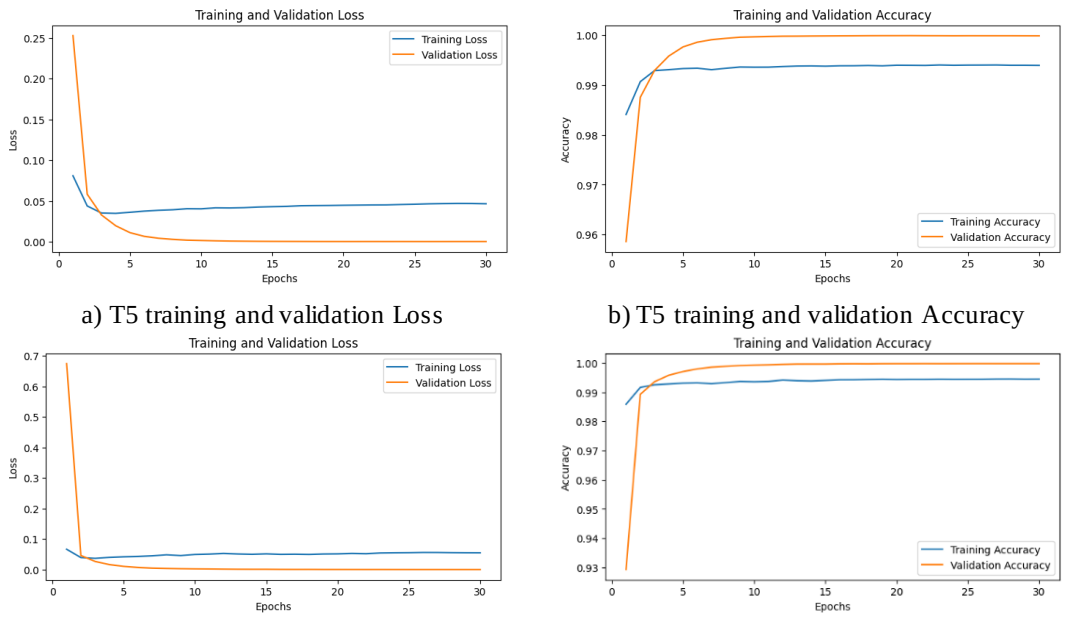
Table 4.3: Model Evaluation Result for CASE I (BART)

Models	Frozen Layers (The lower part)	GLUE	M^2		
			P	R	$F_{0.5}$
BART	2	0.7689	0.7245	0.6088	0.6938
	4	0.7803	0.7835	0.6265	0.7092
	5	0.7798	0.8076	0.6893	0.7618
	6	0.7794	**0.8318**	**0.7521**	**0.8144**
	7	0.7807	0.74205	0.6668	0.7256
	8	**0.7820**	0.6523	0.5815	0.6368
	10	0.7434	0.6649	0.5928	0.6491

The results show that for the BART model, freezing the lower 6 layers achieves GLUE score of 0.7794, with a precision of 0.8318, recall of 0.7521, and an $F_{0.5}$ score of 0.8144. This suggests that the optimal balance between trainable and frozen parameters for BART lies in the 6-layer freezing configuration, which results in a significant improvement in the precision, recall, and $F_{0.5}$ score compared to the other freezing configurations.

For the T5 model, the best performance is observed when freezing 6 layers, with a GLUE score of 0.7679, a precision of 0.8158, a recall of 0.7489, and an $F_{0.5}$ score of 0.8021. While the T5 model's performance is slightly lower than BART's, the trends across the different freezing configurations are similar, indicating that both models benefit from partial layer freezing, with the 6-layer freezing configuration being the most effective.

These baseline results provide a solid foundation for further exploration of the BART and T5 architectures for our specific task and dataset. Figure 4.1 compares the accuracy and loss performance of the T5 and BART models on both the training and validation datasets when the lower 6 encoder and decoder layers are frozen.



a) T5 training and validation Loss

b) T5 training and validation Accuracy

c) BART training and validation Loss

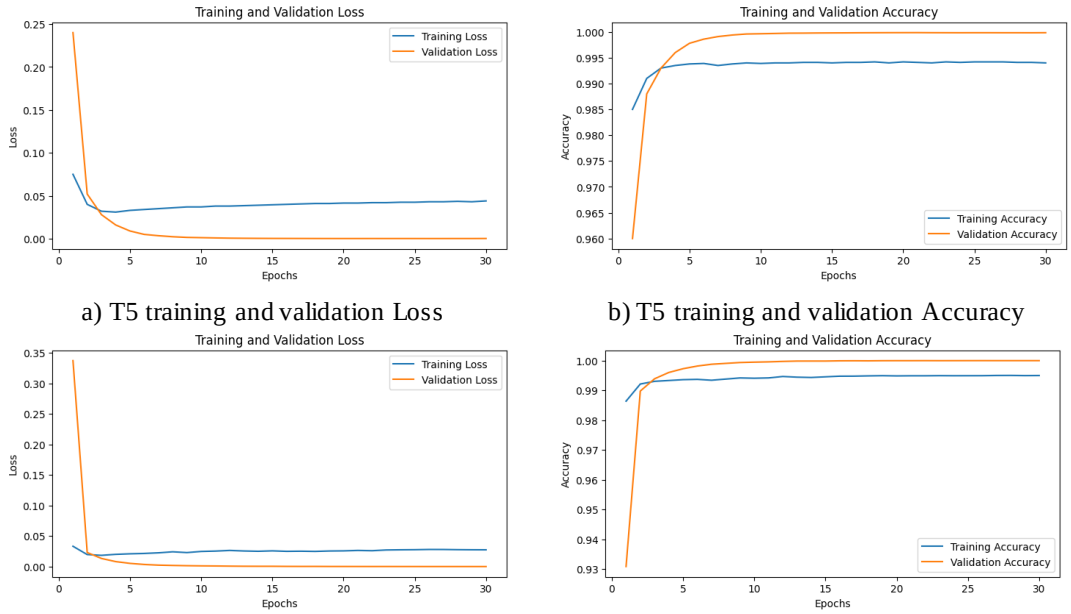
d) BART training and validation Accuracy

Figure 4.1: Training and Validation Results of T5 and BART for CASE I

CASE II: Batch size = 16 and Optimizer = AdamW

The purpose of this case is to explore the effect of batch size. In this case, the models are trained with a batch size of 16 and the AdamW optimizer. Figure 4.2 compares the accuracy and loss of the T5 and BART models on the training and validation sets for this case.

The T5 model has high precision of 0.8125 and recall of 0.7350, resulting in a strong $F_{0.5}$ score of 0.7980. This suggests that the T5 model is effective at identifying and correcting grammatical errors, and is producing high-quality corrections. Additionally, its GLEU score of 0.7658 indicates that the fluency and grammatical correctness of its output is relatively good as presented in the Table 4.4.



a) T5 training and validation Loss

b) T5 training and validation Accuracy

c) BART training and validation Loss

d) BART training and validation Accuracy

Figure 4.2: Training and Validation Results of T5 and BART for CASE II

In contrast, the BART model outperforms T5 in some metrics. BART has a slightly higher precision of 0.8240 and a slightly higher recall of 0.7390, leading to an even stronger $F_{0.5}$ score of 0.8060. This suggests that BART produces high-quality error corrections relatively to T5. However, BART's GLEU score of 0.7738 is only slightly better than T5, indicating that the fluency and grammatical correctness of the two models' outputs are comparable, as shown in the Table 4.3.

The comparison between the batch size 16 and batch size 8 results suggests that increasing the batch size from 8 to 16 has a slightly negative impact on the performance

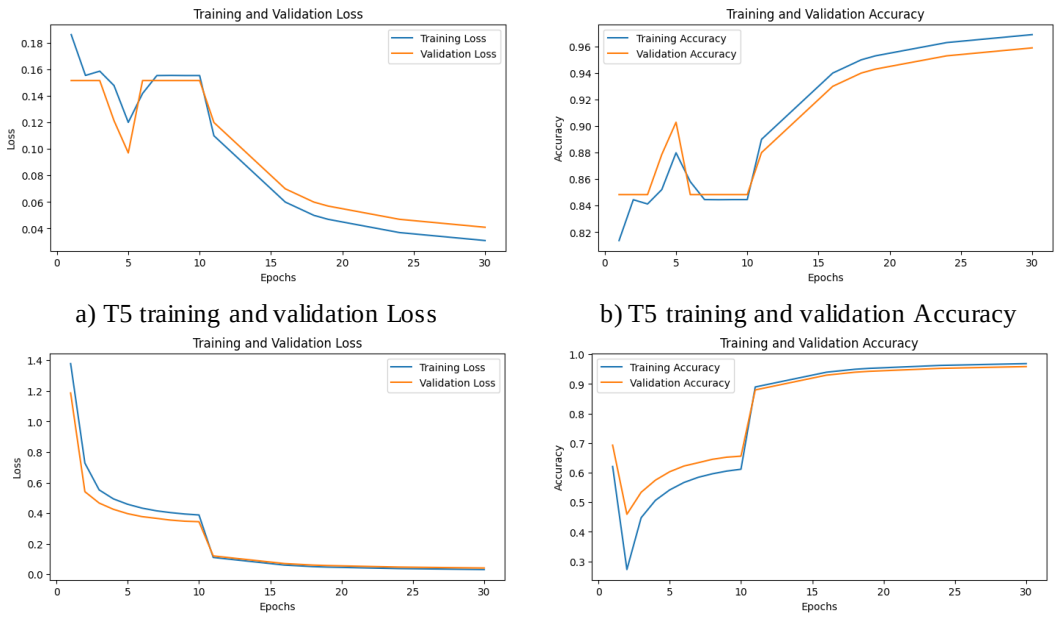
of both T5 and BART models. The precision, recall, $F_{0.5}$, and GLEU scores all decreased slightly for both models, indicating that the smaller batch size of 8 may be more effective in training these GEC models.

Table 4.4: Model Evaluation Result for CASE II

Models	GLEU Score	M^2		
		P	R	$F_{0.5}$
T5	0.7658	0.8125	0.7350	0.7980
BART	**0.7738**	**0.8240**	**0.7390**	**0.8060**

CASE III: Batch size = 8 and Optimizer = RMSProp

In this scenario, the impact of the RMSProp optimizer is investigated. The models are trained using a batch size of 8 and the RMSProp optimizer. Figure 4.3 presents a comparison of the accuracy and loss for the T5 and BART models on both the training and validation datasets in this specific case.



a) T5 training and validation Loss

b) T5 training and validation Accuracy

c) BART training and validation Loss

d) BART training and validation Accuracy

Figure 4.3: Training and Validation Results of T5 and BART for CASE III

The results show that for the BART model, freezing the lower 6 layers achieves GLUE score of 0.7036, with a precision of 0.6386, recall of 0.6034, and an $F_{0.5}$ score of 0.4328. For the T5 model, the best performance is observed when freezing 6 layers, with a GLUE score of 0.6963, a precision of 0.6489, a recall of 0.5876, and an $F_{0.5}$ score of 0.4287 as shown in the Table 4.5.

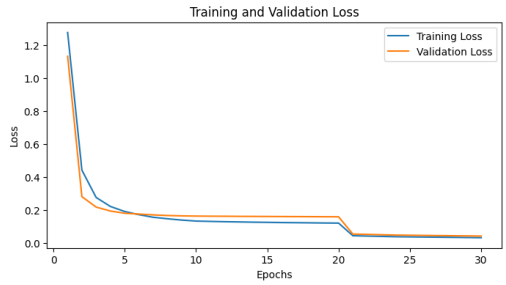
This shows that the choice of optimizer has a significant impact on the performance of the T5 and BART models. With the AdamW optimizer, the models achieve substantially better results across the metrics of Precision, Recall, F0.5, and GLEU Score compared to when using the RMSProp optimizer.

Table 4.5: Model Evaluation Result for CASE III

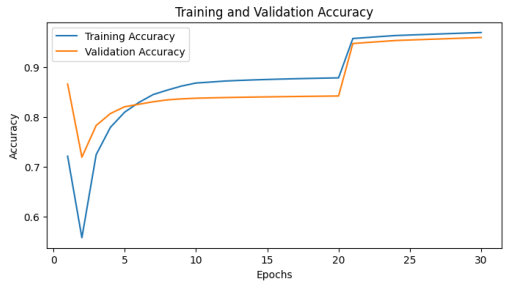
Models	GLEU Score	M^2		
		P	R	$F_{0.5}$
T5	0.6963	**0.6489**	0.5876	0.4287
BART	**0.7036**	0.6386	**0.6034**	**0.4328**

CASE IV: Batch size = 8 and Optimizer = SGD

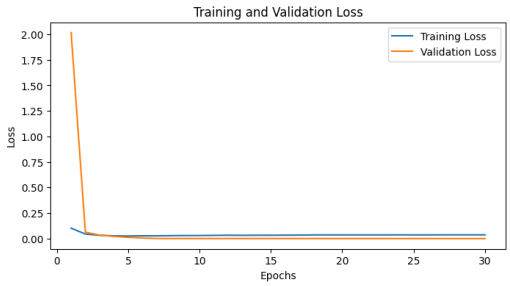
In this context, the effects of the SGD optimizer are examined. The models are trained with a batch size of 8 while utilizing the SGD optimizer. Figure 4.4 illustrates a comparison of the accuracy and loss metrics for the T5 and BART models on the training and validation datasets in this particular instance.



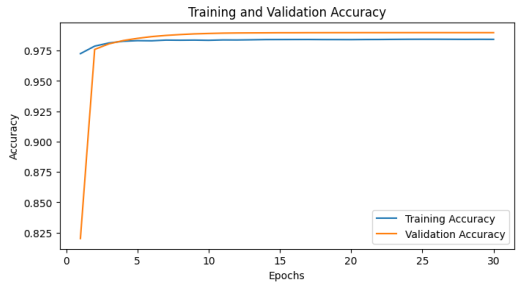
a) T5 training and validation Loss



b) T5 training and validation Accuracy



c) BART training and validation Loss



d) BART training and validation Accuracy

Figure 4.4: Training and Validation Results of T5 and BART for CASE IV

The findings indicate that for the BART model, freezing the bottom 6 layers results in a GLUE score of 0.7135, along with a precision of 0.6869, a recall of 0.5936, and an $F_{0.5}$ score of 0.5097. In the case of the T5 model, the highest performance is achieved

by freezing 6 layers, yielding a GLUE score of 0.6824, a precision of 0.6947, a recall of 0.5863, and an $F_{0.5}$ score of 0.5091 as presented in the Table 4.6.

This indicates when utilizing the AdamW optimizer, the models demonstrate significantly improved outcomes in terms of Precision, Recall, $F_{0.5}$, and GLEU Score compared to the results obtained with the SGD optimizer. These findings imply that the AdamW optimizer is a notably more effective option for these models and tasks than both RMSProp and SGD.

Table 4.6: Model Evaluation Result for CASE IV

Models	GLEU Score	M^2		
		P	R	$F_{0.5}$
T5	0.6824	**0.6947**	0.5863	0.5091
BART	**0.7135**	0.6869	**0.5936**	0.**5097**

4.3 Comparison with Previous Works

While traditional grammar checkers have focused on error detection, GEC aims to provide more comprehensive solutions that also suggest appropriate corrections. The performance of two GEC models, BART and T5, is compared against an existing Hybrid Grammar Checker system for Afaan Oromo in this study. The provided data offers insights into the relative strengths and improvements of the new models compared to the existing system. This data highlights the advancements in NLP and the potential for enhanced grammar checking capabilities.

Table 4.7: Comparison with the Existing System

Methods	Precision	Recall	F-measure
Hybrid Grammar Checker	0.82	**0.76**	0.79
BART (Proposed Model)	**0.8318**	0.7521	**0.8144**
T5 (Proposed Model)	0.8158	0.7489	0.8021

The data in Table 4.7 allows for a comparison between the existing Hybrid Grammar Checker and the proposed solution of BART and T5 models. The Hybrid Grammar Checker system achieves a Precision of 0.82, a Recall of 0.76, and an F-measure of 0.79. This serves as a benchmark for the performance of the existing system.

The proposed models, BART and T5, demonstrate improved performance compared to the existing system. BART achieves a Precision of 0.8318, a Recall of 0.7521, and an $F_{0.5}$ of 0.8087. Similarly, T5 achieves a Precision of approximately 0.8158, a Recall of approximately 0.7521, and an $F_{0.5}$ of 0.8021. The improved Precision and F-measure scores of BART and T5 suggest that these new research models have the potential to outperform the existing system in terms of overall performance. This comparison highlights the advancements made in the field of NLP and the potential of these new transformer models to enhance the capabilities of GEC.

It is important to note that the existing system primarily focused on detecting and suggesting corrections for morphological agreement errors. The existing system also does not provide word order error suggestion. In contrast, the new proposed models, BART and T5, go beyond mere detection and focus on the task of GEC. By not only identifying grammatical errors but also providing accurate and contextually appropriate corrections, BART and T5 offer a more comprehensive solution. This shift in focus from error detection to error correction sets the new models apart from the existing system. The new models are a more valuable tool for users who seek not only to identify grammatical issues but also to improve the overall quality and clarity of their written communication.

4.4 Result Discussion

The experiments conducted on the T5 and BART models demonstrate that both architectures can be effectively adapted to the task of Afaan Oromo GEC. The performance improvements observe as the models are fine-tuned. The potential of transfer learning and fine-tuning as viable strategies for developing robust GEC systems for Afaan Oromo is highlighted.

4.4.1 Impact of Hyperparameters and Optimization algorithms

a. Effect of Number of Epochs

Extending the training duration of the T5 and BART models by increasing the number of epochs leads to substantial improvements in their performance. For the T5 model, key metrics such as precision, recall, $F_{0.5}$ score, and GLEU score show significant enhancements as the number of training epochs is raised from 10 to 20 to 30. Likewise, the BART model exhibits performance improvements across various metrics as the

training is extended for more epochs. This suggests that providing the models with additional training time enables them to learn more effective error correction techniques, ultimately resulting in higher-quality output generation.

b. Effect of Batch Size

Reducing the batch size from 16 to 8 during training leads to a slight improvement in the performance of both the T5 and BART models. For the T5 model, precision increases from 0.8125 to 0.8158, recall improves from 0.7350 to 0.7489, and the $F_{0.5}$ score rises from 0.7980 to 0.8021 when the batch size is halved. Similarly, the BART model exhibits enhancements across all three metrics, with precision increasing from 0.8240 to 0.8318, recall improving from 0.7390 to 0.7521, and the $F_{0.5}$ score rising from 0.8060 to 0.8144. These findings suggest that reducing the batch size during training can lead to modest performance gains for both models, potentially due to the models being exposed to more diverse data samples during each training iteration.

c. Effect of Optimizer

The choice of optimizer also appears to impact the models' performance. For T5, using the AdamW optimizer leads to the best overall results, with high precision, recall, $F_{0.5}$ score, and GLEU score. In contrast, the RMSProp and SGD optimizers result in lower performance across the board. Similarly, for BART, the AdamW optimizer produces the strongest results, while the RMSProp and SGD optimizers lead to lower precision, recall, and $F_{0.5}$ scores. This indicates that the AdamW optimizer is well-suited for training these GEC models and can help them achieve the highest levels of performance.

4.4.2 Research Findings

RQ1: How can a high-quality parallel dataset of Afaan Oromo sentences be developed to evaluate GEC models for this under-resourced language?

A high-quality parallel dataset of Afaan Oromo sentences is developed to evaluate GEC models for this under-resourced language. The final dataset consists of 9,500 parallel Afaan Oromo sentences, where each sentence is paired with its grammatically incorrect and corrected versions. The dataset is sourced from a wide range of topics, including news, social media discussions, educational materials, general conversational language, and various aspects of Afaan Oromo culture. Various types of grammatical errors are included, such as subject-adjective agreement errors, subject-verb agreement errors,

word order errors, adverb-verb agreement errors, and subject and adverb agreement errors. The dataset is carefully compiled and reviewed by native Afaan Oromo speakers and language experts to ensure its accuracy and reliability for GEC models in this under-resourced language.

RQ2: How can existing DL models be adapted for Afaan Oromo GEC using transfer learning, and fine-tuning, considering its unique features and data scarcity?

The results demonstrate that transfer learning and fine-tuning are effective approaches for adapting existing DL models to the task of GEC in the Afaan Oromo language. By leveraging the knowledge and capabilities learned from larger, more resource-rich datasets, the models can be fine-tuned. This allows the models to address the unique linguistic features and data scarcity challenges specific to the Afaan Oromo language.

The performance improvements observed when fine-tuning the T5 and BART models indicate that this approach can be a viable strategy for developing effective GEC solutions for Afaan Oromo, where the availability of annotated data is limited. By building upon the foundations of these pre-trained models, researchers and practitioners can potentially overcome the data scarcity challenge. This allows them to develop more accurate and robust Afaan Oromo GEC systems.

RQ3: Which DL architectures offer the best balance of fluency and overall performance for Afaan Oromo GEC?

Based on the results, the BART model with a batch size of 8 and using the AdamW optimizer appears to offer the best balance of fluency, and overall performance for the Afaan Oromo GEC task, given the constraints of the dataset and resources. The model works well by freezing the 6 lower layers of the encoder and decoder, which helps to improve performance while reducing the number of trainable parameters.

This configuration of the BART model achieved the highest $F_{0.5}$ score of 0.8144, indicating that it is the most accurate among the tested models and configurations. Compared to the T5 model, which reached a maximum $F_{0.5}$ of 0.8021, the BART model demonstrated superior performance.

Regarding fluency, the BART model is generally considered more fluent and natural-sounding in its output compared to the T5 model. This is due to BART's stronger language modeling capabilities and its ability to capture the contextual nuances of the

Afaan Oromo language more effectively. The fluency of the model's output is particularly important for the Afaan Oromo GEC task, as it ensures that the corrected text reads naturally and is easily understood by the end-user.

This section presents the experimental setup, results, evaluation of the T5 and BART models, comparison with previous research, and the study's key findings. In the following section, the study's conclusions are outlined and potential future work is discussed.

CHAPTER 5

CONCLUSION AND FUTURE WORKS

5.1 Conclusion

Grammatical errors can significantly impair communication clarity and effectiveness. This results in decreased reader engagement and hinders the intended message from being conveyed effectively. The primary goal of this work is to develop DL models for effective and robust Afaan Oromo GEC. This research develops GEC for Afaan Oromo using transformer-based models using transfer learning techniques. A dataset of 9,500 parallel Afaan Oromo sentences, each paired with its grammatically incorrect and corrected versions, is prepared and used to train and evaluate the performance of T5 and BART models. The results indicate that the T5 and BART models were highly effective in Afaan Oromo GEC. The BART model achieved a superior $F_{0.5}$ score of 0.8144 compared to 0.8021 for the T5 model. These results demonstrate a performance improvement over the previous Afaan Oromo hybrid grammar checker, with a 3.09% increase for BART and a 1.52% increase for T5. Furthermore, the models were found to work well by freezing the lower layers of the encoder and decoder, which helped improve performance while reducing the number of trainable parameters. These findings show that transfer learning with frozen layers is an effective method for developing high-performing Afaan Oromo GEC models, leveraging the knowledge from pre-trained transformers. This approach can be applied to other low-resource language tasks to achieve strong results with limited data.

5.2 Future works

While the current research has demonstrated the effectiveness of DL models in enhancing GEC for the Afaan Oromo language, there are several promising avenues for future exploration and improvement. One key direction is to expand the dataset used. Increasing the size and diversity of the parallel Afaan Oromo corpus, as well as incorporating more real-world grammatical error samples generated through an automated process, could further enhance the models' performance and robustness. Furthermore, leveraging additional linguistic features or metadata could also contribute to enhancing the models' performance and robustness.

Additionally, investigating other transformer-based architectures may lead to even more accurate and comprehensive GEC. Conducting a more extensive evaluation, including error type analysis, and human evaluation, would provide deeper insights into the models' strengths and limitations.

Finally, a crucial next step is to integrate the DL-based GEC models into real-world applications, such as writing assistants, language learning platforms, or content editing tools. Collaborating with end-users, gathering feedback, and optimizing the models for efficient deployment will ensure the practical impact and long-term sustainability of these technological advancements.

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